SKILL-01

Encapsulation Builder

TCP, UDP and ICMP Packet Capture and Header Analysis

STUDENT DETAILS	INFORMATION
Name	Ramagiri Rishik
Subject	CNNP
Project	Skill01
Experiment	Network Packet Capture and Header Analysis using Wireshark
Roll Number	2520030333
Section	S-1
Date	09-11-2026

Submitted as part of the CNNP Skill work

1. AIM

To capture and analyze network packets using Wireshark, apply display filters for TCP, UDP and ICMP traffic, inspect Ethernet, IPv4, TCP and UDP headers, and calculate packet-header overhead and payload efficiency.

2. SOFTWARE / REQUIREMENTS

- Windows computer with an active network connection
- Wireshark 4.6.9
- Wi-Fi network interface
- Command Prompt for generating ICMP traffic using ping

3. THEORY

Wireshark is a network protocol analyzer used to capture and inspect packets travelling through a network interface. A captured packet can contain several protocol layers.

Protocol / Layer	Purpose
Ethernet II	Carries frames on the local network and contains source and destination MAC addresses.
IPv4	Provides logical source and destination IP addresses and packet information such as header length, total length and TTL.
TCP	Connection-oriented transport protocol. Its header includes source and destination ports, sequence and acknowledgement numbers, and control flags.
UDP	Connectionless transport protocol. Its header contains source port, destination port, length and checksum.
ICMP	Used for network control and diagnostic messages, including ping echo requests and replies.

4. PROCEDURE

1. Open Wireshark and select the active Wi-Fi interface.
2. Double-click the Wi-Fi interface to start packet capture.
3. Generate normal network traffic by opening websites in a browser.
4. Stop the capture and inspect the captured packet list.
5. Apply the display filter tcp to display TCP packets.
6. Apply the display filter udp to display UDP packets.
7. Apply the display filter icmp. If no ICMP packets appear, start a new capture and run ping 8.8.8.8 in Command Prompt. Return to Wireshark and inspect the ICMP packets.
8. Apply the display filter tcp.len > 0 to display TCP packets carrying data.
9. Select a TCP packet and expand Ethernet II, Internet Protocol Version 4 and Transmission Control Protocol in the Packet Details pane.
10. Select a UDP packet and expand User Datagram Protocol to inspect its fields.
11. Save the capture using File > Save As in .pcapng format.

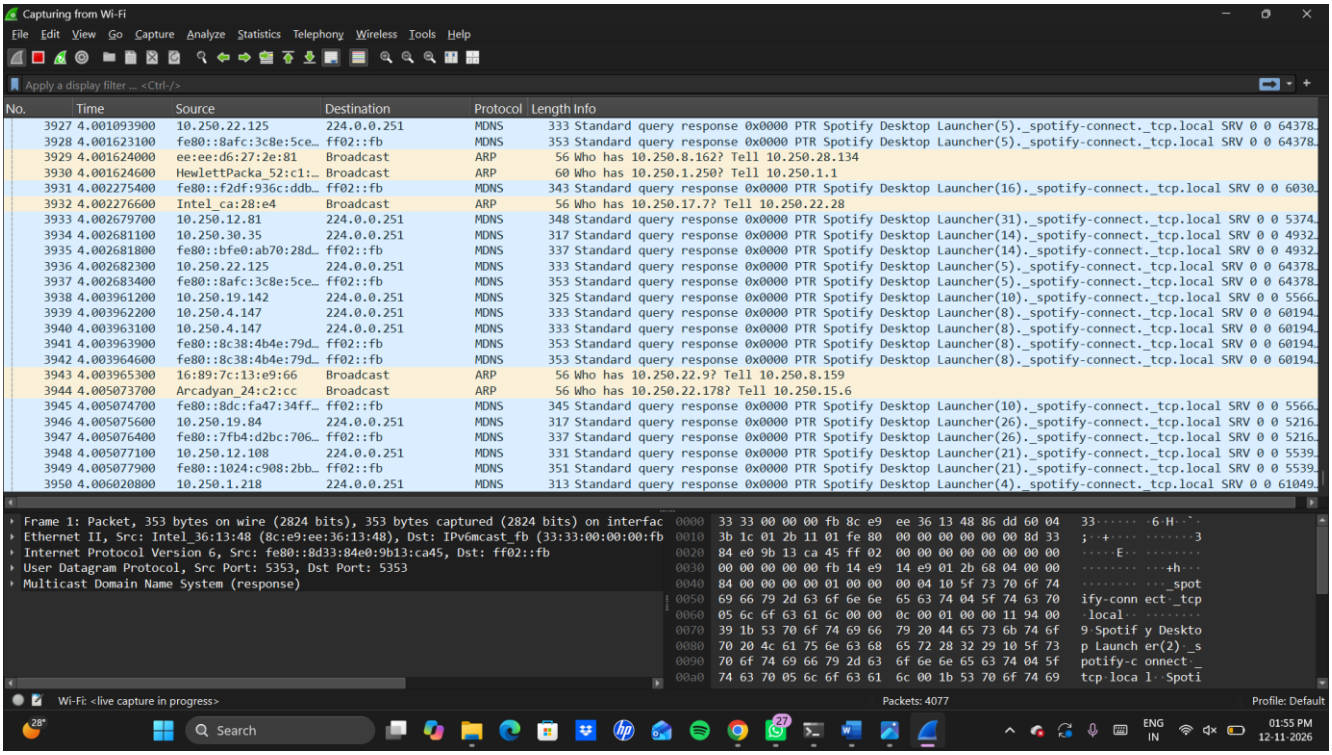
5. DISPLAY FILTERS USED

Filter	Purpose
tcp	Displays TCP packets.
udp	Displays UDP packets.
icmp	Displays IPv4 ICMP packets, such as ping requests and replies.
tcp.len > 0	Displays TCP packets with a non-zero TCP payload.

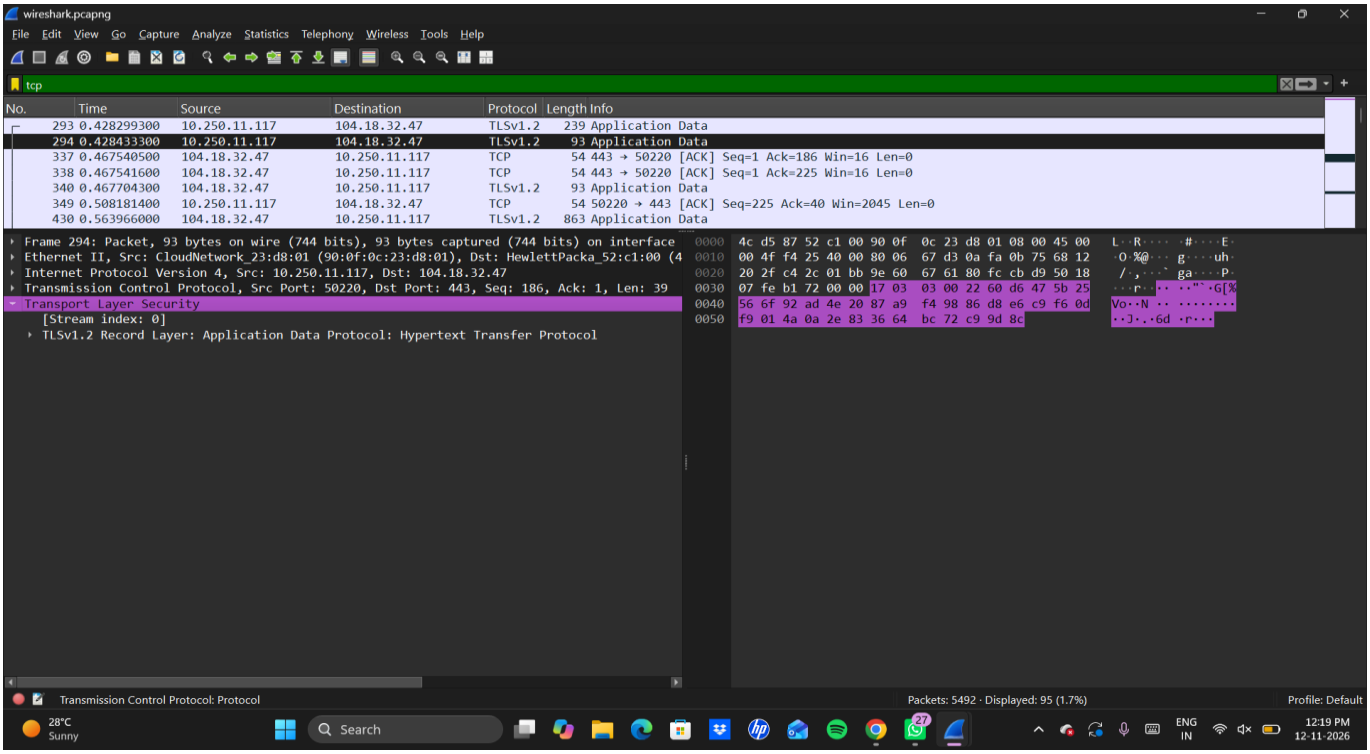
6. SCREENSHOTS / EXPERIMENT EVIDENCE

Paste the corresponding screenshots in the boxes below. Use clear screenshots in which the filter, packet list and relevant packet details are readable.

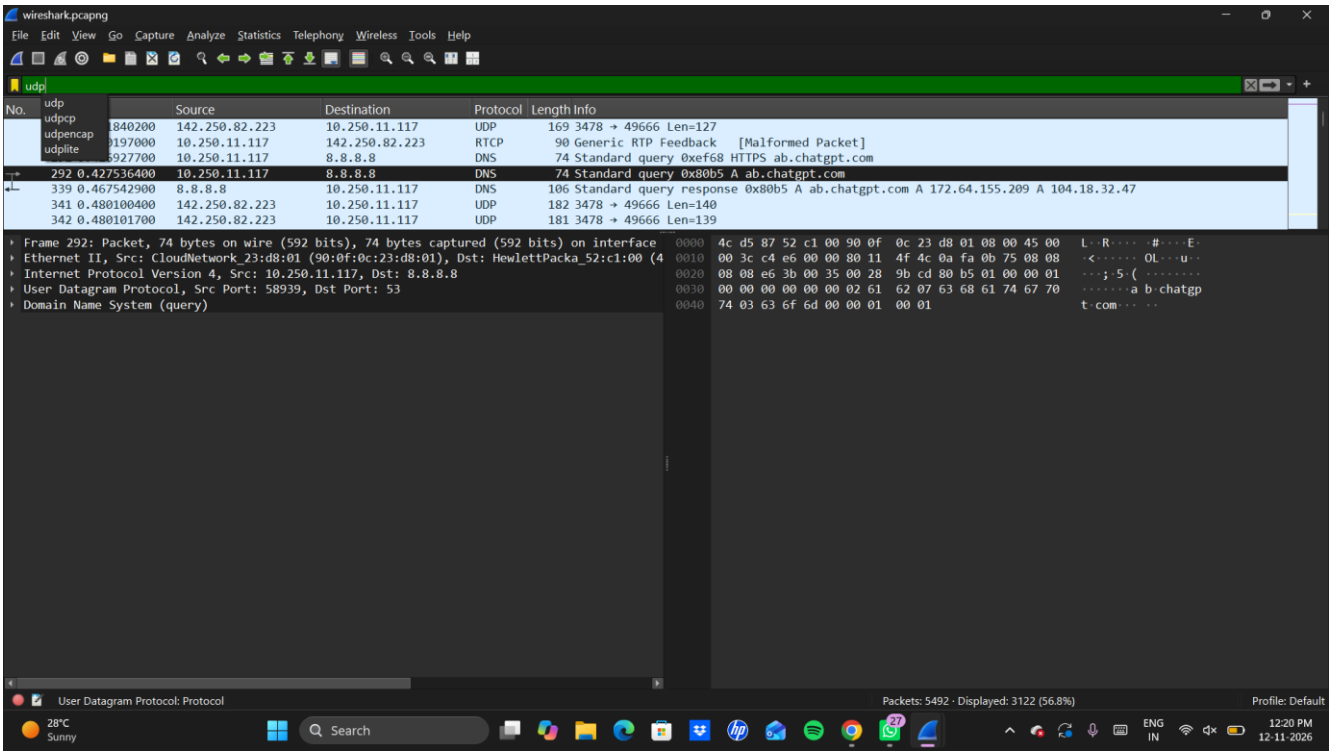
SCREENSHOT 1: Wi-Fi packet capture in progress



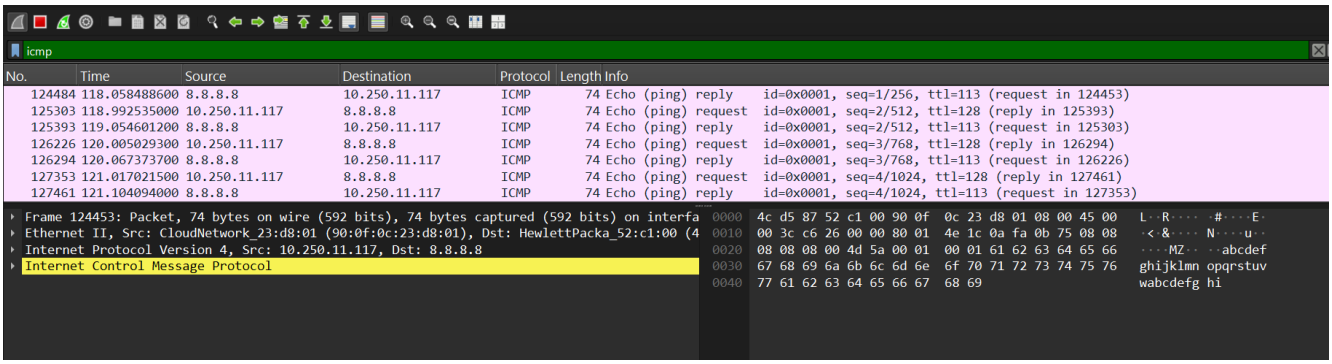
SCREENSHOT 2: TCP display filter



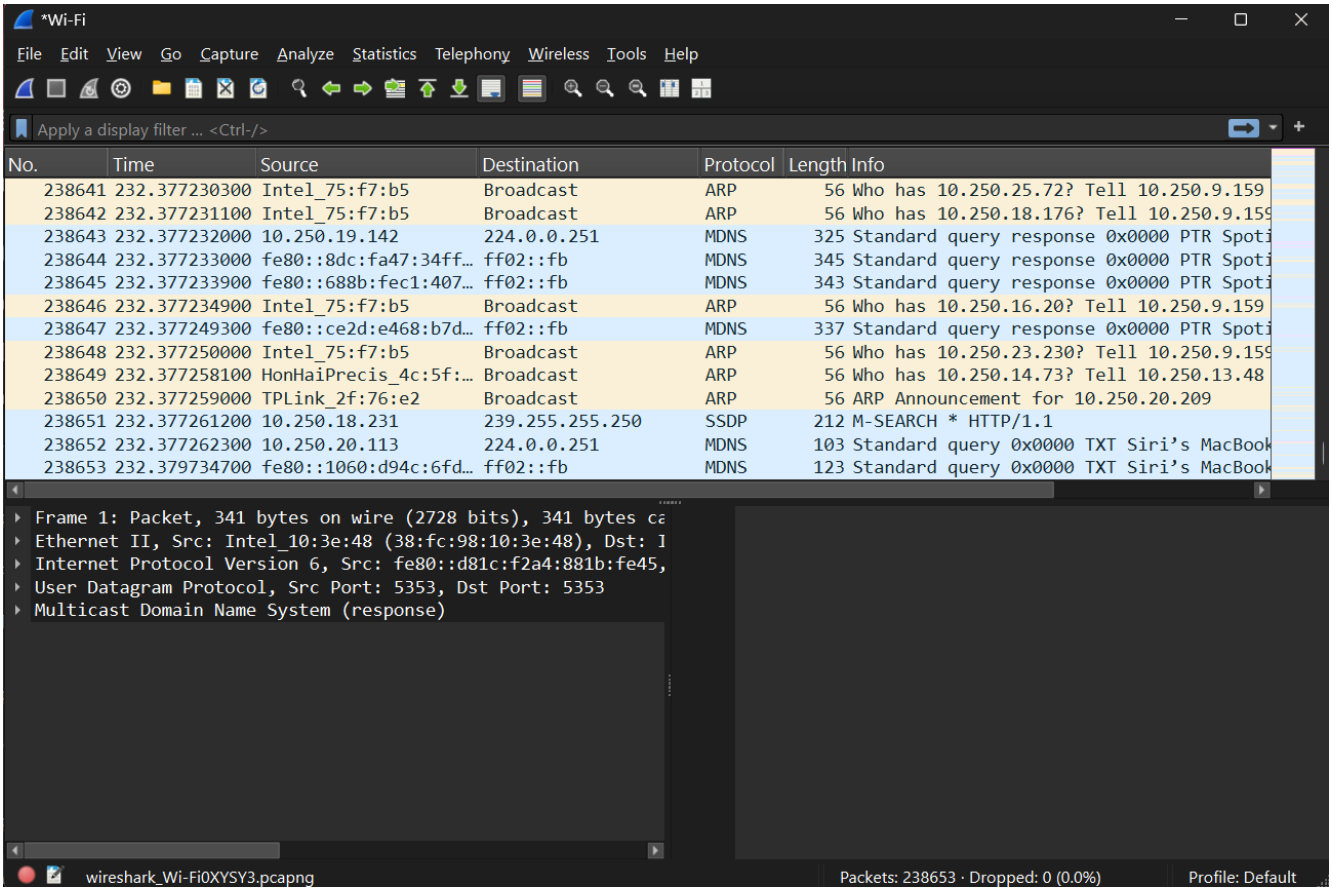
SCREENSHOT 3: UDP display filter



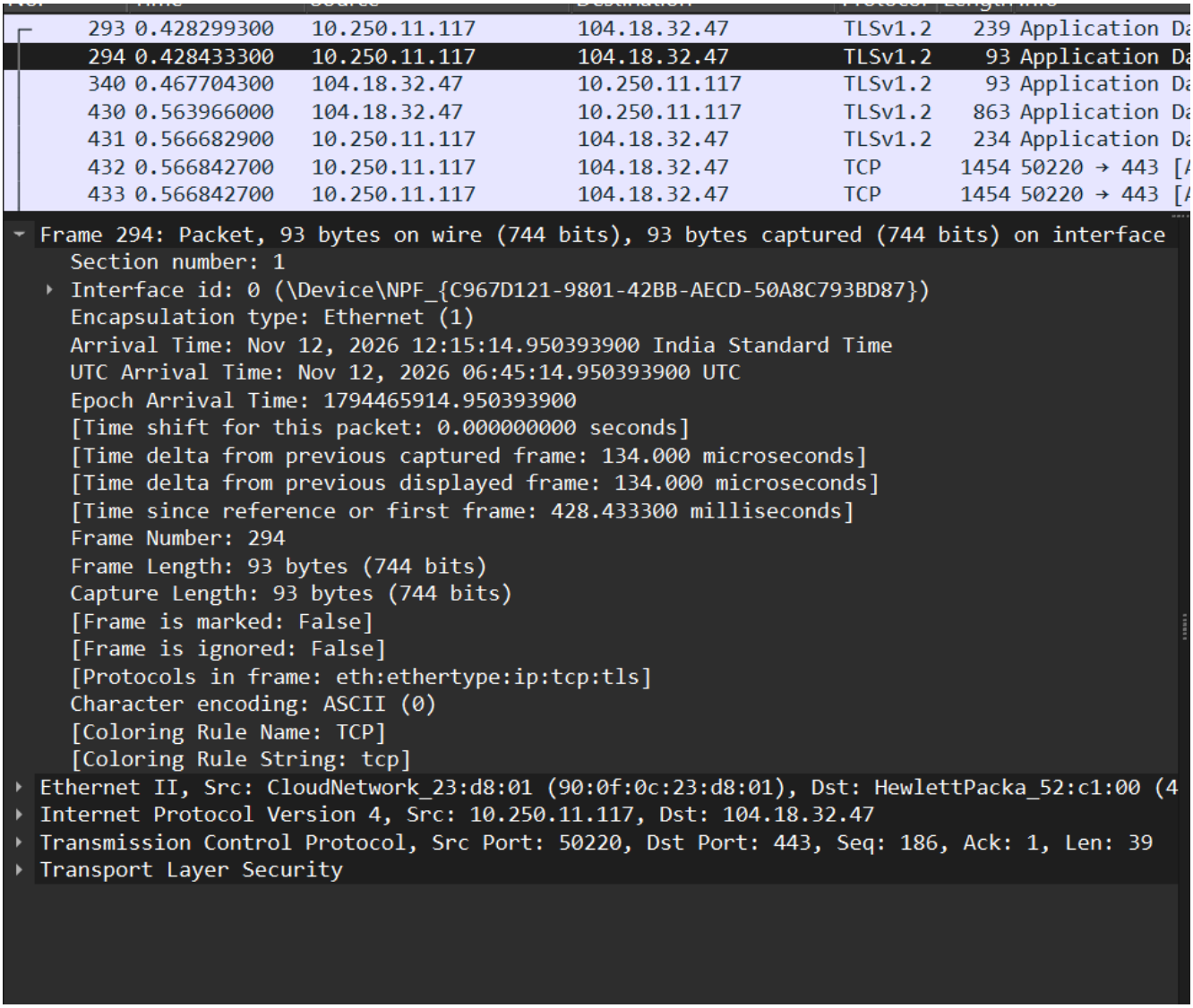
SCREENSHOT 4: ICMP display filter and ping packets



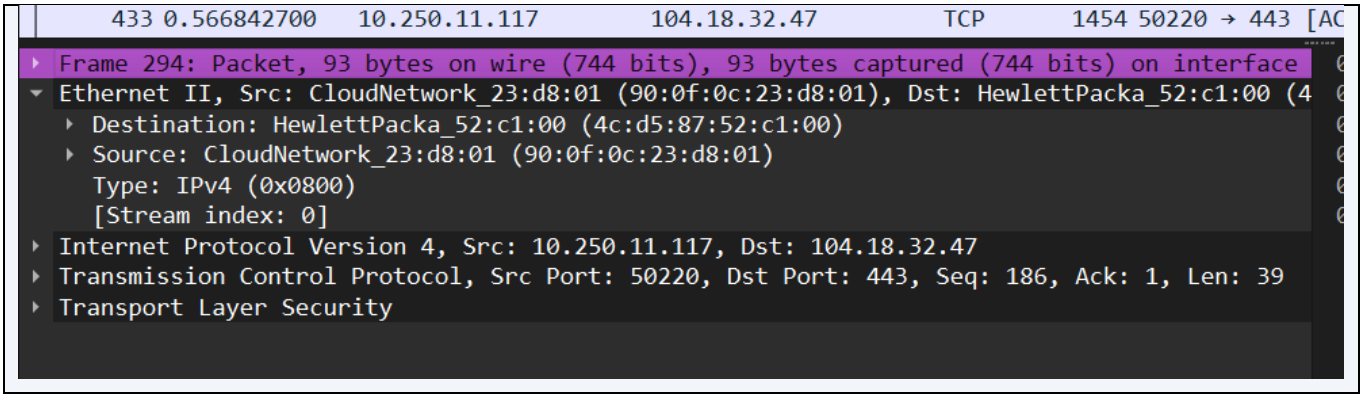
SCREENSHOT 5: TCP packets carrying data



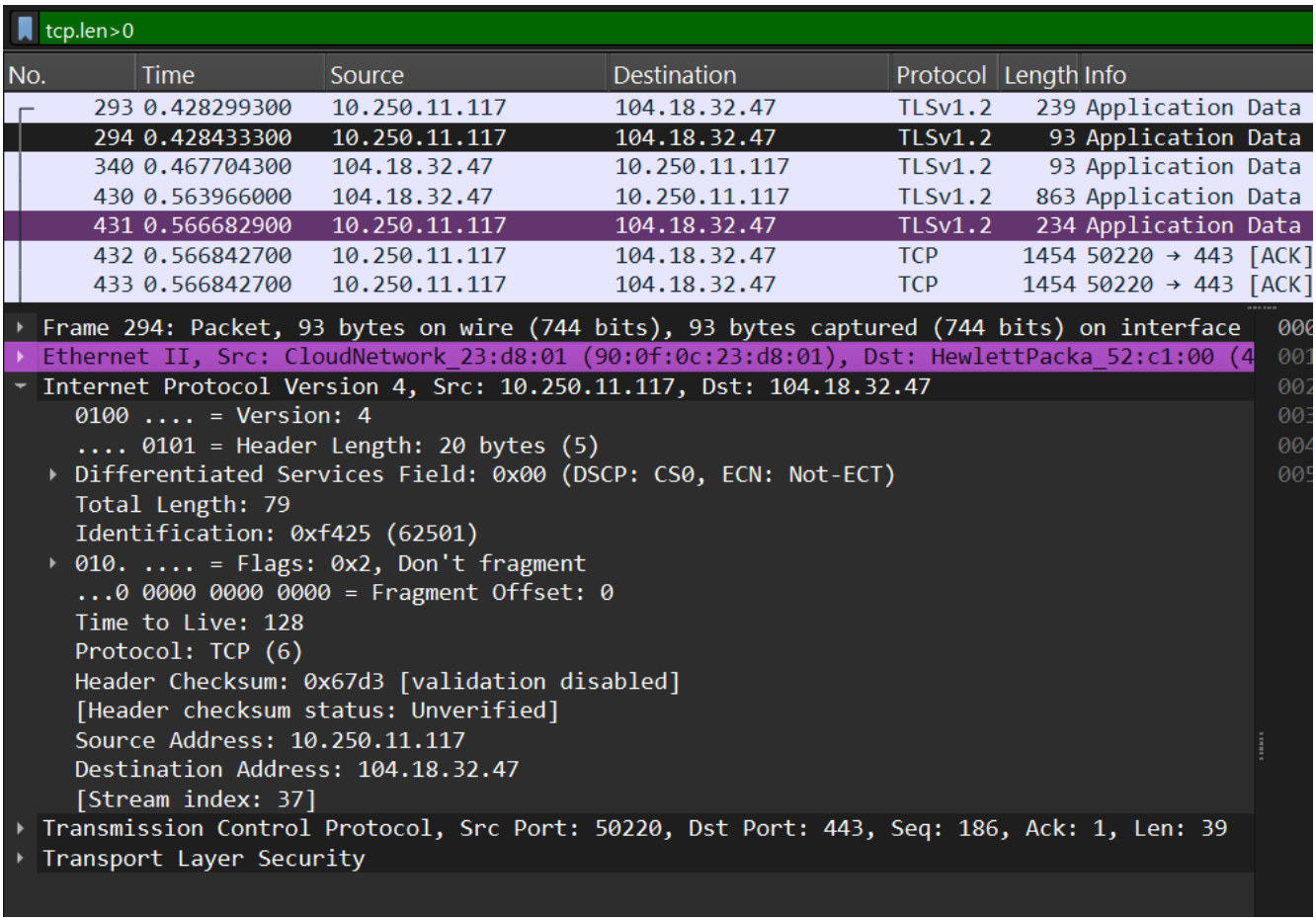
SCREENSHOT 6: Expanded Frame 294



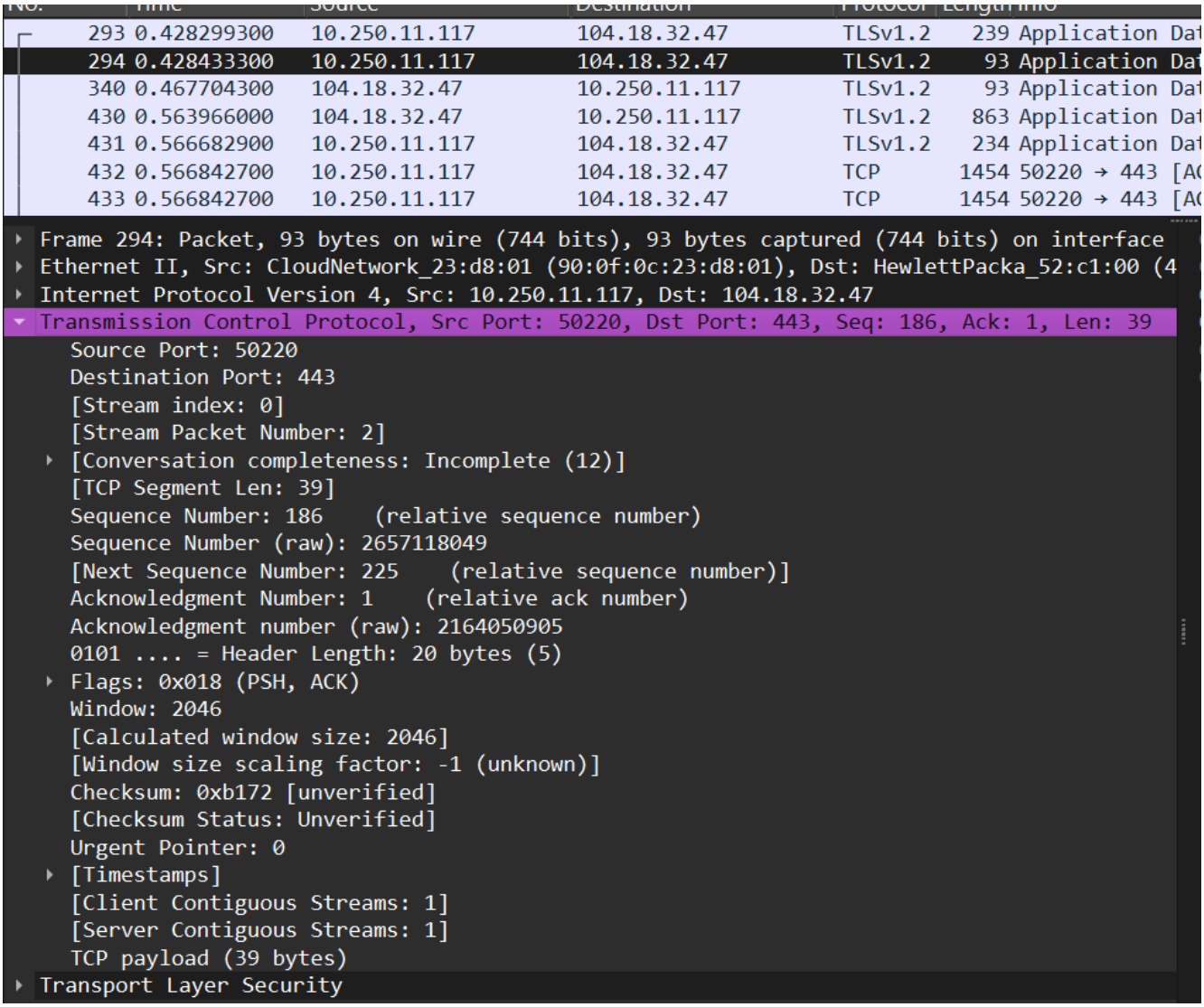
SCREENSHOT 7: Expanded Ethernet II header



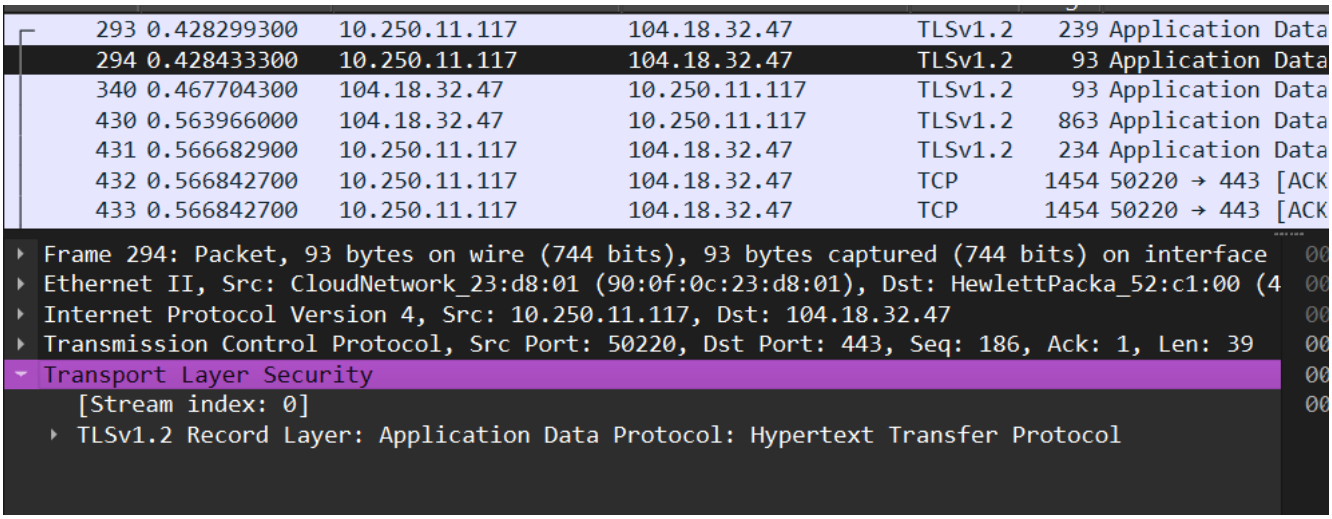
SCREENSHOT 8: Expanded IPv4 section



SCREENSHOT 9: Expanded TCP header



SCREENSHOT 10: Expanded TLS header



7. OBSERVATIONS

7.1 TCP packet observation

The selected TCP packet details recorded during the experiment are shown below.

Parameter	Observed value
Frame length	174 bytes
Ethernet II header	14 bytes
IPv4 header length	20 bytes
IPv4 total length	160 bytes
TCP source port	443
TCP destination port	56393
TCP header length	20 bytes
TCP segment length	120 bytes
TCP payload	120 bytes
Source IP address	149.154.167.99
Destination IP address	10.250.11.117
TTL	41

7.2 UDP packet observation

The selected UDP packet details recorded during the experiment are shown below.

Parameter	Observed value
Frame length	106 bytes
Ethernet II header	14 bytes
IPv4 header length	20 bytes
IPv4 total length	92 bytes
UDP source port	3478
UDP destination port	49666
UDP header length	8 bytes
UDP length	72 bytes
UDP payload	64 bytes
Source IP address	142.250.82.223
Destination IP address	10.250.11.117
TTL	55

7.3 ICMP observation

Parameter	Observed value
Filter	icmp
Source IP (request)	10.250.11.117
Destination IP (request)	8.8.8.8
Packet type	Echo request and Echo reply
Packet length shown	74 bytes
Ping packets sent	4
Ping packets received	4
Packet loss	0%
Average round-trip time	71 ms

8. CALCULATIONS

8.1 TCP overhead and payload efficiency

Total overhead = Ethernet header + IPv4 header + TCP header

= 14 + 20 + 20 = 54 bytes

Total frame length = overhead + TCP payload = 54 + 120 = 174 bytes

Payload efficiency = (TCP payload / total frame length) × 100

= (120 / 174) × 100 = 68.97%

8.2 UDP overhead and payload efficiency

Total overhead = Ethernet header + IPv4 header + UDP header

= 14 + 20 + 8 = 42 bytes

Total frame length = overhead + UDP payload = 42 + 64 = 106 bytes

Payload efficiency = (UDP payload / total frame length) × 100

= (64 / 106) × 100 = 60.38%

Note: These calculations use the captured frame lengths reported by Wireshark. Ethernet FCS was not included in the captured frame lengths.

9. RESULT

Network traffic was successfully captured and analyzed using Wireshark. TCP, UDP and ICMP display filters were applied. Ethernet II, IPv4, TCP and UDP header fields were inspected, and header overhead and payload efficiency were calculated from the selected packets.

10. CONCLUSION

The experiment demonstrated how Wireshark can capture and analyze network packets at different protocol layers. The observations showed the structure of Ethernet, IPv4, TCP and UDP headers and the difference between TCP and UDP packet formats. ICMP ping requests and replies were also captured successfully.

11. VIVA QUESTIONS

What is Wireshark?

A network protocol analyzer used to capture and inspect packets.

What is a display filter?

A filter used to show selected packets from a capture.

What is the filter for TCP?

tcp

What is the filter for UDP?

udp

What is the filter for ICMP?

icmp

What is the TCP header size in the observed packet?

20 bytes.

What is the UDP header size?

8 bytes.

What is the purpose of ICMP ping?

To test reachability and measure round-trip time.

What is a MAC address?

A hardware address used for communication on a local network.

What does TTL mean?

Time To Live; it limits how long an IP packet can travel through routers.